

LECTURE-53 Z-TRANSFORM & ROC



Introduction

- Just like Laplace Transform is a broader version of CTFT
Z-Transform is a broader version of DTFT.
- Z-Transform is discrete counter part of Laplace Transform.
- just like LT converts differential equations to algebraic equation
Z-Transform also converts difference equation to algebraic equation.

Types of Sequences

(1) Right sided Sequence:

if samples of the sequence exist to
right of some $n = n_0$

i.e. $x[n] = 0 \quad \forall n < n_0$

duration of sequence = ∞

$$\text{eg } u[n+3] \quad [-3, \infty)$$

(2) Left sided sequence:

if samples of the sequence exist to
left of some $n = n_0$

i.e. $x[n] = 0 \quad \forall n > n_0$

duration = ∞

$$u[-n+3] \quad (-\infty, 3]$$

(3) Both sided sequence:

if samples extend to ∞ on both sides of $n=0$

i.e. sequence exists from $n=-\infty$ to $n=\infty$

then sequence is said to be Both Sided Sequence.

$$a^{|n|}, \quad 0 < a < 1 \quad (-\infty, \infty)$$

(4) Finite Duration sequence:

If the number of samples in the sequence are finite then it is said to be finite duration

sequence.

$$u[n+3] - u[n-2] \quad [-3, 1]$$

Z-Domain

- After taking z-transform the independent variable becomes z instead of n & it is said to z-domain.

(1) Rational Function

$$X(z) = \frac{N(z)}{D(z)}$$

where $N(z)$ & $D(z)$ are polynomials in z

order of num^r < order of deno^r : proper function

order of num^r $\geq$ order of deno^r : improper function

(2) Irrational Function

if either numerator or denominator is not a polynomial in z then it is said to be **Irrational function**.

eg: $\frac{\sin z}{z+1}$

Poles and Zeros of z-domain signal

- roots of numerator i.e. values of z at which $\text{num}^r(N(z))$ becomes 0 or $X(z)$ becomes 0 are said to be zeroes of $X(z)$.
- roots of denominator i.e. values of z at which $\text{deno}^r(D(z))$ becomes 0 or $X(z)$ becomes ∞ are called as poles of $X(z)$.

$$X(z) = \frac{K(z-z_1)(z-z_2) \dots (z-z_m)}{(z-p_1)(z-p_2) \dots (z-p_n)}$$

$z_1, z_2, \dots, z_m$: zeroes of $X(z)$

$p_1, p_2, \dots, p_n$: poles of $X(z)$

Z-Plane

z-plane is a complex 2D plane

x-axis: $\text{Re}(z)$

y-axis: $\text{Im}(z)$

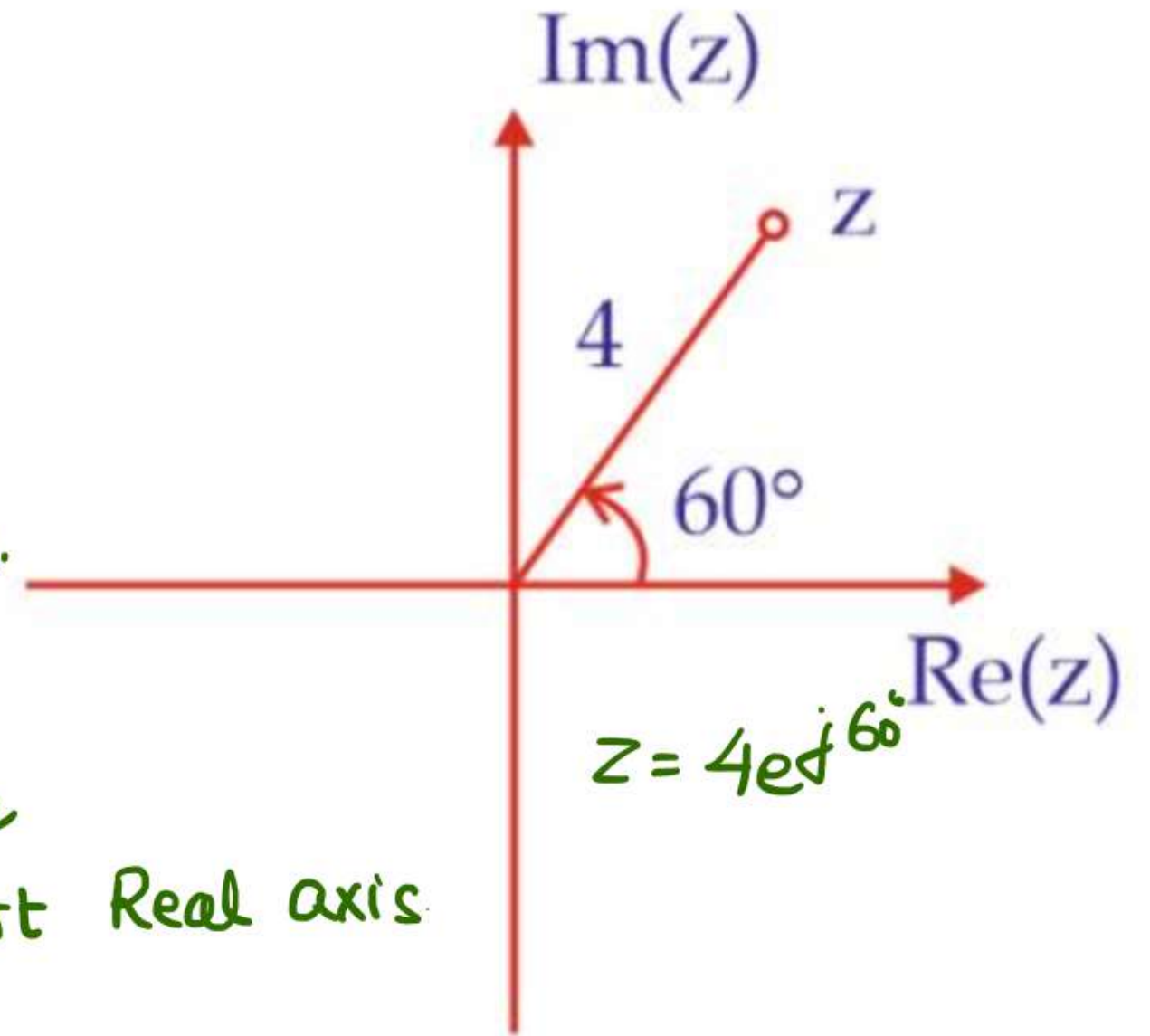
Values of z are taking in polar form.

$$z = r e^{j\Omega}$$

$|z| = r$: distance of point from origin

Ω : angle made by posⁿ vector wrt Real axis

$$0 \leq r < \infty \quad 0 \leq \Omega < 2\pi$$



Types of Z-transform

(1) Bilateral Z-Transform

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$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

- Can be used for both causal & non-causal signals.
- Initial condⁿ are considered separately as additional input.
- It can describe freq response & stability.

(2) Unilateral Z-Transform

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

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SN

- use of UZT is restricted to causal functions
- considers initial condⁿ during solution of difference equations.

Region of Convergence (ROC)

- ROC is a region in z -plane that contains all such values of z where $|X(z)| = \text{finite}$
- ROC make z -transform of a signal unique.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$|X(z)| = \left| \sum_{n=-\infty}^{\infty} x[n] z^{-n} \right|$$
$$\leq \sum_{n=-\infty}^{\infty} |x[n] (re^{j\omega})^{-n}|$$

$$|X(z)| \leq \sum_{n=-\infty}^{\infty} |x[n] r^{-n}|$$

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∴ for $X(z)$ to be finite

$$\sum_{n=-\infty}^{\infty} |x[n] r^{-n}| = \text{finite}$$

i.e. $x[n] r^{-n}$: absolutely summable

Values of ' r ' for which $x[n] r^{-n}$ is absolutely summable lie in ROC

ROC is in the form of circles centered at origin.

Properties of ROC

- ① ROC of z-transform contains rings centered at origin in z-plane.
- ② ROC does not contain any pole.
- ③ if $x[n]$ is of finite duration then ROC is entire z-plane except possible $z=0$ or $z \rightarrow \infty$ ** SN
- ④ if $x[n]$ is a right sided sequence & if $|z| = r_0$ is a part of ROC then all values of z s.t. $|z| > r_0$ lie in ROC.
- ⑤ if $x[n]$ is a left sided seq. & $|z| = r_0 \in \text{ROC}$ then all values of z s.t. $|z| < r_0$ lie in ROC.
- ⑥ if $x[n]$ is right sided, ROC is outside outermost pole **
- ⑦ if $x[n]$ is left sided, ROC is inside innermost pole. **
- ⑧ if $x[n]$ is both sided, ROC is annular region b/w 2 poles. **

Mapping of s-plane to z-plane

• z-transform can be thought of as Laplace transform of sampled version of $x(t)$

$$x(t) \rightarrow x[n] = x(nT)$$

T : sampling interval

$$x[n] = x(t) \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \delta(t-nT)$$

$$X(s) = \sum_{n=-\infty}^{\infty} x(nT) e^{-snT}$$

$$x(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$X(s) = \sum_{n=-\infty}^{\infty} x[n] (e^{sT})^{-n}$$

on comparing $z = e^{sT}$

$$z = e^{(\sigma + j\omega)T} = e^{\sigma T} \cdot e^{j\omega T}$$

$$r e^{j\Omega} = e^{\sigma T} \cdot e^{j\omega T}$$

$$r = e^{\sigma T} \quad \Omega = \omega T \text{ radians}$$

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Case-1: left half plane ($\sigma < 0$)

$$r = e^{\sigma T} < 1$$

∴ it is mapped to interior of unit circle.

Case-2: Imaginary axis ($\sigma = 0$)

$$r = e^{\sigma T} = 1$$

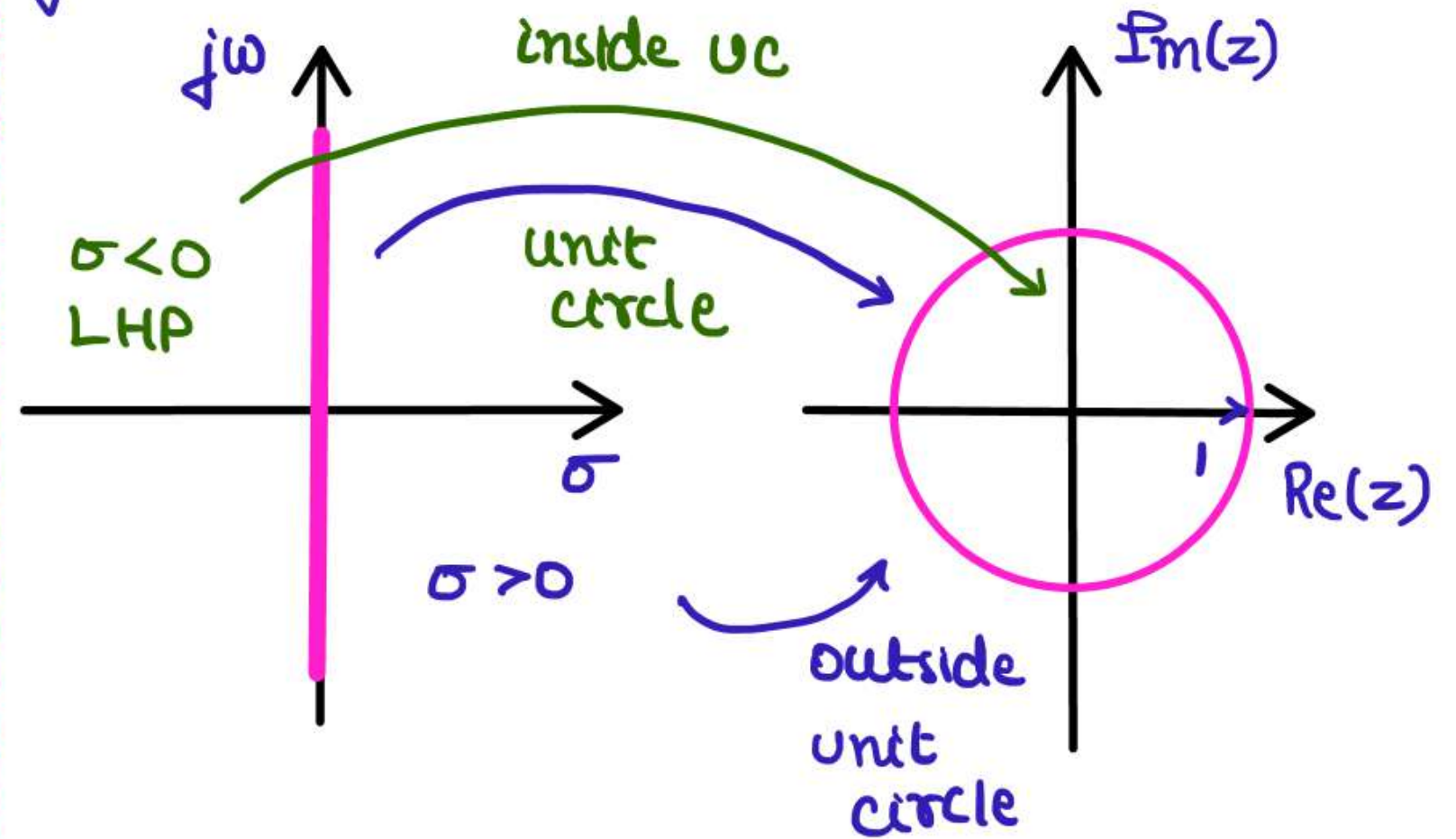
boundary of unit circle

Case-3: right half plane ($\sigma > 0$)

$$r = e^{\sigma T} > 1$$

outside unit circle

~~SN~~ Vertical strips of ROC in s-domain get converted to circles in z-domain.



Question-01

Find z-transform of $x[n] = a^n u[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

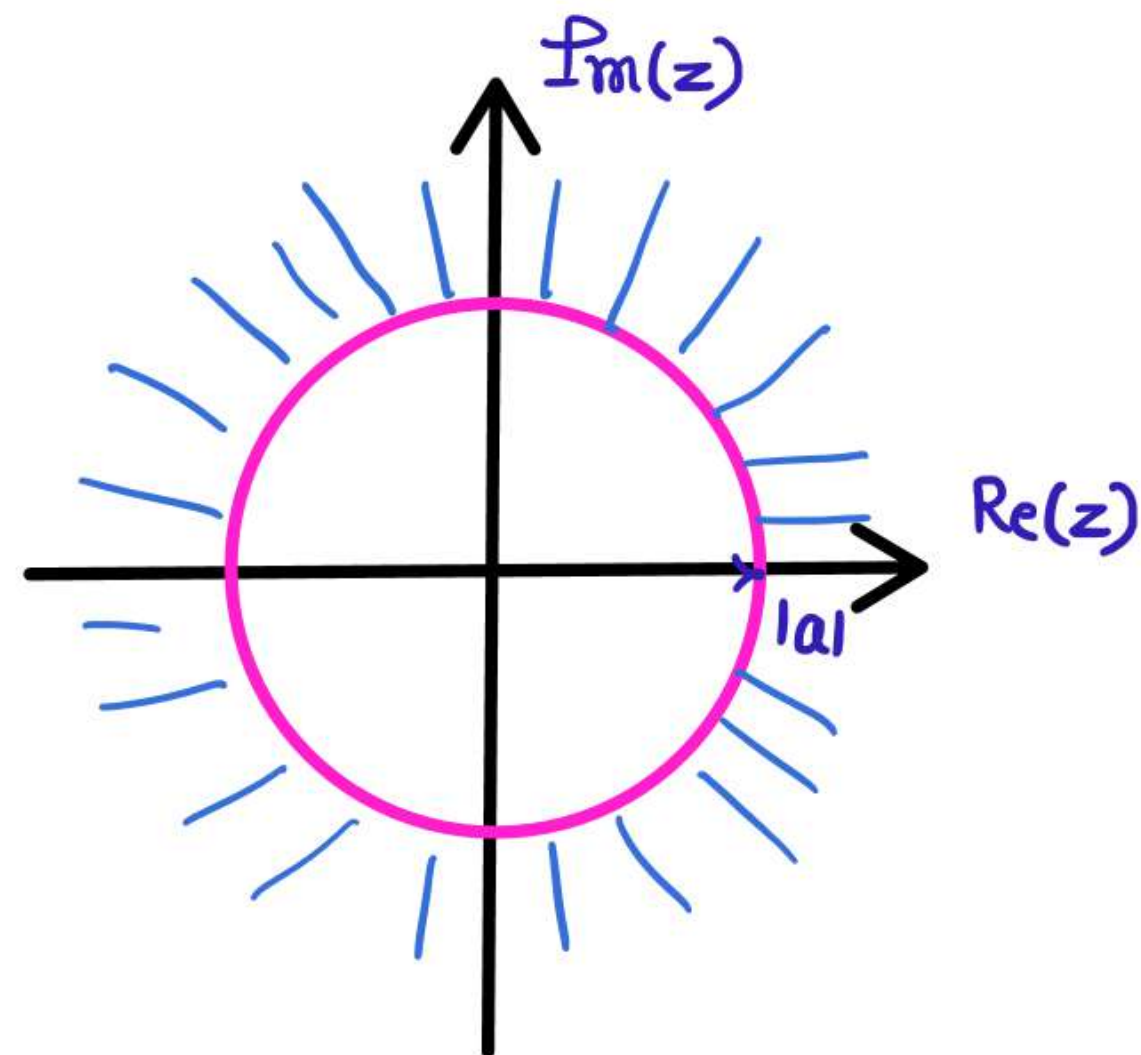
$$= \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n$$

sum of infinite GP

$$X(z) = \frac{1}{1 - a/z} \quad ; \quad \text{cond}^n \left| \frac{a}{z} \right| < 1$$
$$|z| > |a|$$

$$= \frac{1}{1 - az^{-1}} \quad ;$$

$$= \frac{z}{z - a}$$



Question-02

Find z-Transform of $x[n] = -a^n u[-n-1]$?

$\hookrightarrow (-\infty, -1]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -a^n z^{-n}$$

$$= \sum_{n=-\infty}^{-1} -\left(\frac{a}{z}\right)^n$$

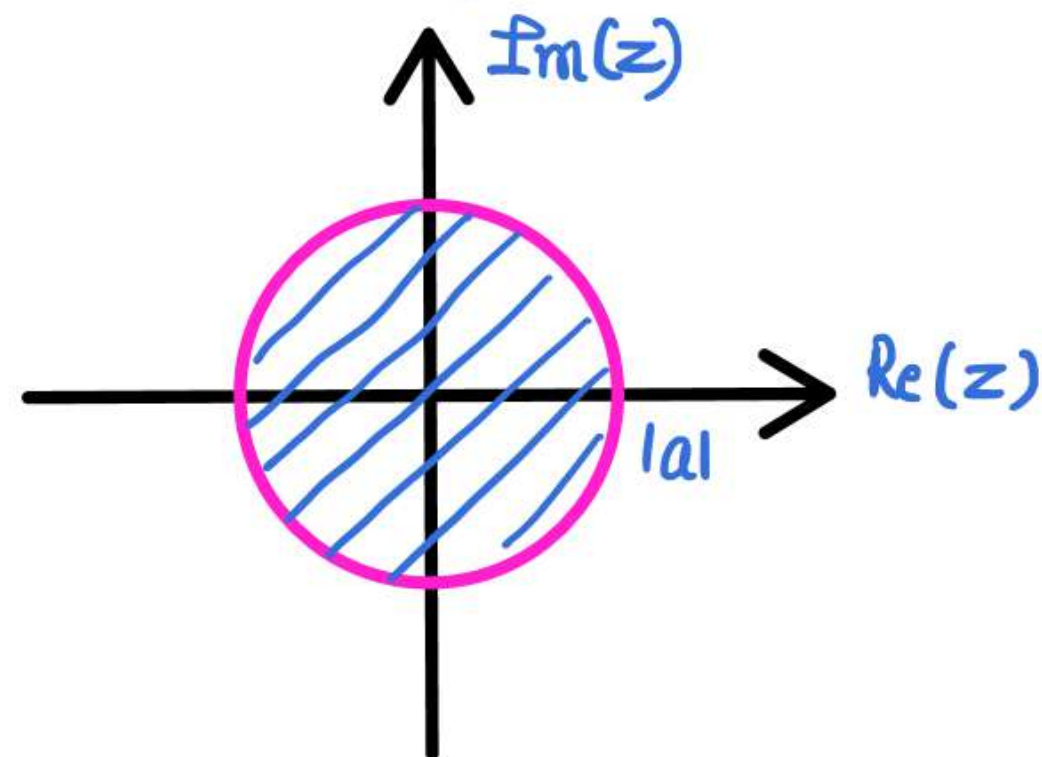
$$n = -k$$

$$X(z) = \sum_{k=1}^{\infty} -\left(\frac{a}{z}\right)^{-k} = \sum_{k=1}^{\infty} -\left(\frac{z}{a}\right)^k$$

$$X(z) = \frac{-z/a}{1 - z/a} = \frac{-z}{a-z}$$

$$= \frac{z}{z-a} = \frac{1}{1-az^{-1}}$$

$$\text{cond}^n \quad \left|\frac{z}{a}\right| < 1, \quad |z| < |a|$$



Important point

$$a^n u[n] \xrightarrow{ZT} \frac{z}{z-a} ; |z| > |a|$$

$$-a^n u[-n-1] \xrightarrow{ZT} \frac{z}{z-a} ; |z| < |a|$$

$u[-n-1]$ is analogous of $u(-t)$

$$e^{-at} u(t) \rightarrow \frac{1}{s+a} ; \sigma > -a$$

$$-e^{-at} u(-t) \rightarrow \frac{1}{s+a} ; \sigma < -a$$

Question-03

Determine Z-Transform and ROC of following signals

(i) $x[n] = \delta(n)$

$$(i) \quad X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

(ii) $x[n] = \{1, 2, 3, \}$
 $\uparrow$

$$= \sum_{n=-\infty}^{\infty} \delta[n] z^{-n} = z^{-n} \Big|_{n=0} = 1$$

ROC: entire z -plane

(ii) $x[n] = \{1, 2, 3\}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} = x[0]z^{-0} + x[1]z^{-1} + x[2]z^{-2}$$
$$= 1 + \frac{2}{z} + \frac{3}{z^2}$$

ROC: entire z plane except $z=0$

Question-04

Determine Z-Transform and ROC of following signals

(i) $x[n] = \{3, 4, 5, \}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

(ii) $x[n] = (6 + j8)^n u[n]$

$$= x[-1]z + x[0]z^{-0} + x[1]z^{-1}$$

$$= 3z + 4 + \frac{5}{z}$$

ROC: entire z-plane
except $z=0$ & $z=\infty$

(iii) $a^n u[n] \rightarrow \frac{z}{z-a} ; |z| > |a|$

$$= \frac{z}{z-(6+j8)} ; |z| > |6+j8|$$
$$|z| > 10$$

Question-05

Determine Z-Transform and ROC of following signals

(i) $x[n] = (3 + j4)^n u[-n-1]$ $-a^n u[-n-1] \rightarrow \frac{z}{z-a} \quad |z| < |a|$

$$(ii) \quad x[n] = 2^n u[n] + 3^{-n} u[n] \quad \rightarrow \quad (1/3)^n$$

$$(ii) \quad \frac{z}{z-2} + \frac{z}{z-\frac{1}{3}}$$

$$= \frac{z}{z-2} + \frac{z}{z^{-1/3}} ; |z| > 2$$

right sided signal : outside
outermost pole

Question-06

Determine Z-Transform and ROC of following signals

(i) $x[n] = 2^n u[-n-1] + 3^{-n} u[n]$

(i) $\frac{-z}{z-2} + \frac{z}{z-\frac{1}{3}} ; \frac{1}{3} < |z| < 2$

(ii) $x[n] = 2^n u[-n-1] - 3^{-n} u[-n-1]$

$|z| < 2 \quad |z| > \frac{1}{3}$

both sided signal

ROC $\rightarrow$ annular region b/w 2 poles

$\downarrow$
 $\frac{-z}{z-2} + \frac{z}{z-\frac{1}{3}}$

$|z| < 2 \quad |z| < \frac{1}{3}$

ROC: $|z| < \frac{1}{3}$

left sided signal : ROC $\rightarrow$ inside inner most pole

Question-07

Determine Z-Transform and ROC of following signals

(i) $x[n] = 2^n u[n] - 3^{-n} u[-n-1]$

$$X(z) = \frac{z}{z-2} + \frac{z}{z-\frac{1}{3}}$$

$$|z| > 2 \quad |z| < \frac{1}{3}$$

ROC does not exist

Z-Transform does not exist

(ii) $x[n] = \begin{cases} -\left(\frac{1}{2}\right)^n, & n < 0 \\ 0, & n \geq 0 \end{cases}$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1]$$

$$= \frac{z}{z-\frac{1}{2}}, \quad |z| < \frac{1}{2}$$

Question-08

later using property

Consider the D.E.

$$y[n] - \frac{1}{3}y[n-1] = x[n] = \times[n] \text{ and } x[n] = \left(\frac{1}{2}\right)^n u[n]. \quad \underline{\underline{HW}}$$

Assuming the condition of initial rest, the solution for $y(n)$: $n \geq 0$ is

(a) $3\left(\frac{1}{3}\right)^n - 2\left(\frac{1}{2}\right)^n$

(b) $-2\left(\frac{1}{3}\right)^n + 3\left(\frac{1}{2}\right)^n$

(c) $\frac{2}{3}\left(\frac{1}{3}\right)^n + \frac{2}{3}\left(\frac{1}{2}\right)^n$

(d) $\frac{1}{3}\left(\frac{1}{3}\right)^n + \frac{2}{3}\left(\frac{1}{2}\right)^n$

Question-09

The 'z' transform $X(z)$ of the signal

$$x[n] = \left(-\frac{1}{4}\right)^n u[-n-3] \text{ is } \rightarrow (-\infty, -3]$$

(a) $X[z] = \frac{-64z^3}{1+4z} ; |z| < \frac{1}{4}$

(b) $X[z] = \frac{64z^3}{1-4z} ; |z| < \frac{1}{4}$

(c) $X[z] = \frac{-64z^2}{1+4z^2} ; |z| < \frac{1}{4}$

(d) $X[z] = \frac{-64z^3}{1+4z^2} ; |z| < \frac{1}{4}$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{-3} \left(-\frac{1}{4}\right)^n z^{-n}$$

$$= \sum_{n=-\infty}^{-3} \left(-\frac{1}{4}z\right)^n$$

assume $n = -k$

$$X(z) = \sum_{k=3}^{\infty} \left(-\frac{1}{4}z\right)^{-k} = \sum_{k=3}^{\infty} (-4z)^k$$

$$= \frac{(-4z)^3}{1+4z} ; |4z| < 1$$
$$|z| < \frac{1}{4}$$

Question-10

Consider three different signals

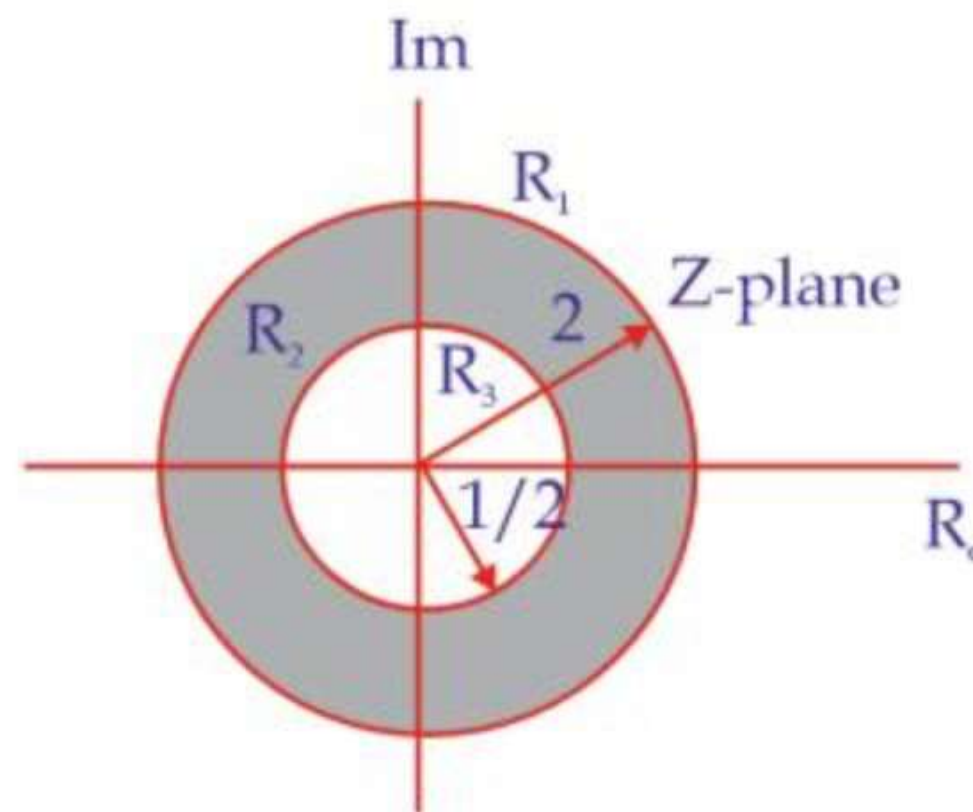
$$x_1[n] = \left[2^n - \left(\frac{1}{2}\right)^n \right] u[n] \quad \text{right sided}$$

$$x_2[n] = -2^n u[-n-1] + \left(\frac{1}{2}\right)^n u[-n-1] \quad \text{left sided}$$

$$x_3[n] = -2^n u[-n-1] - \left(\frac{1}{2}\right)^n u[n] \quad \text{both sided}$$

Following figure shows the three different regions. Choose the correct for the ROC of signal

R_1	R_2	R_3
(a) $x_1[n]$ ✓	$x_2[n]$	$x_3[n]$
(b) $x_2[n]$	$x_3[n]$	$x_1[n]$
☒ (c) $x_1[n]$	$x_3[n]$	$x_2[n]$
(d) $x_3[n]$	$x_2[n]$	$x_1[n]$



Question-11

$H(z)$ is transfer function of discrete system and has two poles at $z = \frac{1}{2}$ & 2

$H(z)$ is rational. ROC includes $z = 3/4$, $h(0) = 1$ and $h(-1) = 4$.

Impulse response $h(n)$ of the system is

(a) $h(n) = \left(\frac{1}{2}\right)^n u(n) + 2^{n+2} u(n-1)$

(b) $h(n) = -\left(\frac{1}{2}\right)^n u(n) - 2^{n+2} u(-n-1)$

(c) $h(n) = \left(\frac{1}{2}\right)^n u(n) + 2^{n+3} u(-n-1)$

(d) $h(n) = -\left(\frac{1}{2}\right)^n u(n) - 2^{n+2} u(-n)$

$c_1 \left(\frac{1}{2}\right)^n u[n]$
 $+ c_2 (2)^n u[-n-1]$
 $\rightarrow h[0] = -1$

$h[0] = 1$

